

Statement of intent

Baha Khammari · PhD position, auction-based support schemes for renewable energy
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I am applying for the doctoral position on auction-based support schemes for renewable energy in Europe. I want to explain what I would bring to it, and where I would need to build.

Where I come to this from

My energy training is at Cologne, where the Institute of Energy Economics teaches the specialisation. I took *Energy Markets and Regulation* with Jun.-Prof. Dr. Oliver Ruhnau, whose chair is in energy market design, and finished it with 1.3. That course covered electricity market design, network regulation, and the instruments used to bring renewable generation into a market that was not built for it. I then took *Growth, Energy and Climate Change* with PD Dr. Dietmar Lindenberger.

My specialisation seminar paper is on the **EU Emissions Trading System**. Writing it left me with the question that runs through this project as well: a policy instrument is a set of rules, the rules determine who bids what and on what terms, and the aggregate outcome is a consequence of that design rather than of the policy target. An auction for renewable capacity and a cap-and-trade scheme are both attempts to buy an environmental good through a market, and both succeed or fail on the detail.

What I would work on

Of the four directions in the call, the two that I would most want to take up are **pre-qualification requirements and completion bonds** and **local content requirements**. My reason is specific.

Auction design in Europe is not a single policy. Member states adopted auctions at different dates, and then changed the rules at different dates again: penalty levels, bid bonds, realisation deadlines, technology-specific versus technology-neutral lots, and local content conditions have all been introduced and revised on different national timetables. That is staggered adoption of a treatment whose intensity varies, which is the setting my master's thesis was built for.

My thesis estimated the labour-market effects of a national scholarship programme across 558 regions over fifteen years, using the Callaway and Sant'Anna (2021) heterogeneity-robust staggered difference-in-differences estimator on a panel of roughly five million linked administrative records that I constructed myself. The central finding was methodological as much as substantive: the conventional two-way fixed effects estimate was not merely imprecise, it had the wrong sign, because the comparisons it makes between later-treated and already-treated units are invalid once effects vary across groups and over time. It was graded 1.0.

I would like to bring that to auction design. My reading is that the theory of multi-unit auctions for renewables is well developed and that credible causal evidence on what specific design features do is thinner, partly because the variation is staggered and therefore easy to analyse badly. A project that pairs auction theory with a design that can survive that variation is one I would be glad to spend four years on, and it is the contribution I think I could make that a purely theoretical candidate could not.

Where I would need to build, said plainly

The call asks for graduate training in microeconomic theory, industrial organization and game theory. My graduate coursework is econometric rather than theoretical: my strongest marks are in econometrics, mathematics and energy regulation, and my microeconomics grade is average. I would be starting the theory side of this project from a lower base than a candidate trained in IO, and I would rather say so than let the transcript say it for me.

What I would offer against that is that the gap is the kind that closes with coursework and reading, and that the direction it closes in is one I have already chosen. I read Mas-Colell and Osborne and Rubinstein as the standard I am expected to reach, and I would treat the first year as the year in which I reach it. The empirical craft, by contrast, is not something a four-year programme usually has time to teach, and I already have it.

Practicalities

I complete my M.Sc. in Economics at Cologne this month and am available from October 2026, so the autumn start suits me. I work in English, hold IELTS Academic 8.0, and speak German and Tunisian Arabic natively and French at B1. I hold EU citizenship, so there is no work-permit question.

My referees are **Prof. Dr. Pia Pinger**, who chaired my master's thesis, and **Jun.-Prof. Dr. Oliver Ruhnau**, who taught and examined me in energy market design. Contact details are on the CV. I am glad to send the thesis in full, and the seminar paper on the EU ETS.

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